

Homework Set #5

S4.2.3

$$\psi'(x) = e^{i/\hbar p_0 x} \psi(x)$$

$$\langle \psi' | P | \psi' \rangle = \int dx \psi'(x)^* \frac{\hbar}{i} \frac{d}{dx} \psi'(x)$$

$$= \int dx \psi'(x)^* \frac{\hbar}{i} \frac{d}{dx} e^{i/\hbar p_0 x} \psi(x)$$

$$= \int dx \psi'(x)^* e^{i/\hbar p_0 x} \left(p_0 + \frac{\hbar}{i} \frac{d}{dx} \right) \psi(x)$$

$$= \int dx \psi(x)^* \left(p_0 + \frac{\hbar}{i} \frac{d}{dx} \right) \psi(x)$$

$$= p_0 + \langle \psi | P | \psi \rangle$$

S5.1.1

$$1 = \int_{-\infty}^{+\infty} dp |p\rangle \langle p|$$

$$= \int_{-\infty}^0 dp |E = \frac{p^2}{2m}, -\rangle \langle E = \frac{p^2}{2m}, -|$$

$$+ \int_0^{\infty} dp |E = \frac{p^2}{2m}, +\rangle \langle E = \frac{p^2}{2m}, +|$$

$$= \int_0^{\infty} dE \left| \frac{dp}{dE} \right| \sum_{\epsilon=\pm} |E, \epsilon\rangle \langle E, \epsilon|$$

$$= \sum_{\epsilon=\pm} \int_0^{\infty} dE \sqrt{\frac{m}{2E}} |E, \epsilon\rangle \langle E, \epsilon|$$

55.1.2

$$\langle x | H | E \rangle = E \langle x | E \rangle$$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi_E(x) = E \psi_E(x)$$

The most general solution is

$$\psi_E(x) = A_+ e^{+\frac{i}{\hbar} \sqrt{2mE} x} + A_- e^{-\frac{i}{\hbar} \sqrt{2mE} x}$$

$$\langle x | E \rangle = \beta \langle x | E, + \rangle + \gamma \langle x | E, - \rangle$$

$$\beta = A_+ \sqrt{2\pi\hbar}$$

$$\gamma = A_- \sqrt{2\pi\hbar}$$

$$|E\rangle = \beta |E, +\rangle + \gamma |E, -\rangle$$

S5.1.3

$$U(t) \psi(x, 0) = \exp\left(\frac{i}{\hbar} t \frac{\hbar^2}{2m} \frac{d^2}{dx^2}\right) \frac{1}{\pi^{1/4}} e^{-x^2/2}$$

$$= \frac{1}{\pi^{1/4}} \sum_{l=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{m! k!} \left(\frac{i}{\hbar} t \frac{\hbar^2}{2m}\right)^k \left(\frac{-1}{2}\right)^k \frac{d^{2l}}{dx^{2l}} x^{2k}$$

$$\frac{d^p}{dx^p} x^q = 0 \quad \text{if } p > q$$

$$= q(q-1)\dots(q-p+1) x^{q-p} \quad \text{if } p \leq q$$

$$= \frac{1}{\pi^{1/4}} \sum_{l=0}^{\infty} \sum_{k=l}^{\infty} \frac{1}{l! k!} \left(\frac{it\hbar}{2m}\right)^l \left(\frac{-1}{2}\right)^k \frac{(2k)!}{(2k-2l)!} x^{2(k-l)}$$

$$m = k - l \quad k = l + m$$

$$= \frac{1}{\pi^{1/4}} \sum_{m=0}^{\infty} \sum_{l=0}^{\infty} \frac{(2l+2m)!}{l! (l+m)! (2m)!} \left(\frac{-1}{2}\right)^m \left(\frac{-it\hbar}{4m}\right)^l x^{2m}$$

$$\frac{(2p)!}{p!} = \frac{(2p)(2p-1)(2p-2) \dots 2 \cdot 1}{p \cdot (p-1) \dots 1}$$

$$= 2^p (2p-1)(2p-3) \dots 1$$

$$= 2^{2p} \left(p - \frac{1}{2}\right) \left(p - \frac{3}{2}\right) \dots \frac{3}{2} \cdot \frac{1}{2}$$

$$U(t) \psi(x, 0) = \frac{1}{\pi^{1/4}} \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} \left(\frac{-1}{2}\right)^n$$

$$\times \sum_{l=0}^{\infty} \frac{1}{l!} \left(\frac{-i\hbar k}{4m}\right)^l 2^{2(l+n)} \frac{1}{2} \cdot \frac{3}{2} \dots \left(l+n - \frac{1}{2}\right)$$

$$= \frac{1}{\pi^{1/4}} \sum_{n=0}^{\infty} \frac{1}{n! \cancel{2^{2n}} \frac{1}{2} \cdot \frac{3}{2} \dots \left(n - \frac{1}{2}\right)} x^{2n} \left(\frac{-1}{2}\right)^n 2^{2n}$$

$$\times \sum_{l=0}^{\infty} \frac{1}{l!} \left(\frac{-i\hbar k}{m}\right)^l \frac{1}{2} \cdot \frac{3}{2} \dots \left(l+n - \frac{1}{2}\right)$$

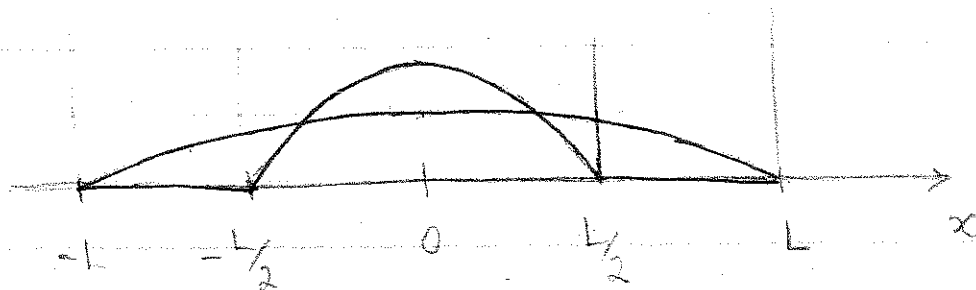
$$U(t) \psi(x, 0) = \frac{1}{\pi^{1/4}} \sum_{n=0}^{\infty} \frac{1}{n!} x^{2n} \left(\frac{-1}{2}\right)^n$$

$$\left[1 + \sum_{l=1}^{\infty} \frac{1}{l!} \left(\frac{-itk}{m}\right)^l (n + \frac{1}{2})(n + \frac{3}{2}) \dots (l + n - \frac{1}{2}) \right]$$

$$= \frac{1}{\pi^{1/4}} \sum_{n=0}^{\infty} \frac{1}{n!} x^{2n} \left(\frac{-1}{2}\right)^n \left(1 + \frac{itk}{m}\right)^{-n - \frac{1}{2}}$$

$$= \frac{1}{\sqrt{\pi} \left(1 + \frac{itk}{m}\right)^{1/2}} \exp\left[-\frac{1}{2} \cdot \frac{x^2}{\left(1 + \frac{itk}{m}\right)}\right]$$

S.B.2.1



$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L}\left(x + \frac{L}{2}\right)\right) \quad \text{for } |x| \leq \frac{L}{2}$$

$$= 0 \quad \text{for } \frac{L}{2} \leq |x| \leq L$$

$$\psi'(x) = \sqrt{\frac{1}{L}} \sin\left(\frac{\pi}{2L}(x + L)\right)$$

$$p = |\langle \psi' | \psi \rangle|^2 = \left| \int_{-L}^{+L} dx \, \psi'(x)^* \psi(x) \right|^2$$

$$= \left(\sqrt{\frac{2}{L}} \int_{-L/2}^{L/2} dx \, \sin\left(\frac{\pi}{L}\left(x + \frac{L}{2}\right)\right) \sin\left(\frac{\pi}{2L}(x + L)\right) \right)^2$$

$$= \left(\frac{\sqrt{2}}{L} \int_0^L dx' \, \sin\left(\frac{\pi x'}{L}\right) \sin\left(\frac{\pi}{2L}\left(x' + \frac{L}{2}\right)\right) \right)^2$$

$$p = \left(\frac{\sqrt{2}}{L} \int_0^{\pi} d\theta \frac{L}{\pi} \sin \theta \sin \left(\frac{1}{2} \theta + \frac{\pi}{4} \right) \right)^2$$

$$\theta = \frac{\pi x'}{L}$$

$$p = \left(\frac{\sqrt{2}}{\pi} \int_0^{\pi} d\theta \frac{1}{2} \left[\cos \left(\frac{1}{2} \theta - \frac{\pi}{4} \right) - \cos \left(\frac{3\theta}{2} + \frac{\pi}{4} \right) \right] \right)^2$$

$$= \left(\frac{1}{\sqrt{2}\pi} \left[2 \sin \left(\frac{1}{2} \theta - \frac{\pi}{4} \right) - \frac{2}{3} \sin \left(\frac{3\theta}{2} + \frac{\pi}{4} \right) \right] \right)^2$$

$$= \left[\frac{1}{\sqrt{2}\pi} \left(2 \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) - \frac{2}{3} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) \right) \right]^2$$

$$= \left[\frac{2}{\pi} \left(1 + \frac{1}{3} \right) \right]^2 = \left(\frac{8}{3\pi} \right)^2$$

S5.2.2 a

$$H|E_n\rangle = E_n|E_n\rangle$$

$$\langle E_n|E_{n'}\rangle = \delta_{nn'}$$

$$\sum_{n=0}^{\infty} |E_n\rangle\langle E_n| = 1$$

E_0 is the
lowest
energy
eigenvalue

$$H = H \sum_{n=0}^{\infty} |E_n\rangle\langle E_n|$$

$$= \sum_{n=0}^{\infty} E_n |E_n\rangle\langle E_n|$$

$$\langle \psi | H | \psi \rangle = \sum_{n=0}^{\infty} E_n \langle \psi | E_n \rangle \langle E_n | \psi \rangle$$

$$= \sum_{n=0}^{\infty} E_n |\langle \psi | E_n \rangle|^2$$

$$= E_0 + \sum_{n=0}^{\infty} (E_n - E_0) |\langle \psi | E_n \rangle|^2$$

≥ 0

≥ 0

since $\sum_{n=0}^{\infty} |\langle \psi | E_n \rangle|^2 = \langle \psi | \psi \rangle = 1$

Note: $\langle \psi | H | \psi \rangle = E_0$ only if $|\psi\rangle = |E_0\rangle$

b.

$$\psi_\alpha(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

$$\int_{-\infty}^{+\infty} dx |\psi_\alpha(x)|^2 = \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} e^{-\alpha x^2} dx = 1$$

$$V(x) = -|V(x)| \quad \text{for all } x$$

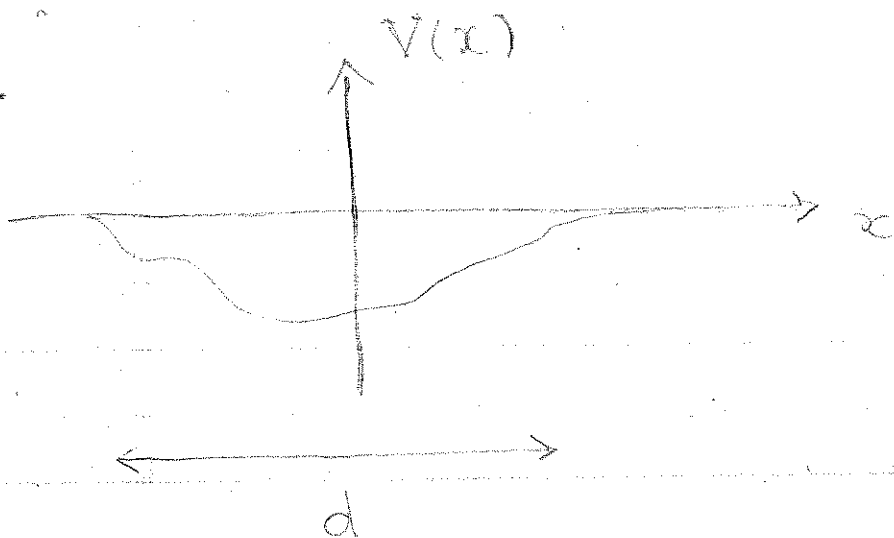
$$E_\alpha = \langle \psi_\alpha | H | \psi_\alpha \rangle$$

$$= \int_{-\infty}^{+\infty} dx \sqrt{\frac{\alpha}{\pi}} e^{-\alpha x^2/2} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - |V(x)| \right) e^{-\alpha x^2/2}$$

$$= \frac{1}{2m} \langle \psi_\alpha | P^2 | \psi_\alpha \rangle - \langle \psi_\alpha | |V(x)| | \psi_\alpha \rangle$$

$$\langle \psi_\alpha | P^2 | \psi_\alpha \rangle = \frac{\hbar^2}{2 \Delta^2} \Big|_{\Delta = \frac{1}{\sqrt{\alpha}}} = \frac{\hbar^2 \alpha}{2}$$

since $\psi_\alpha(x)$ is a Gaussian



$$\langle \Psi_\alpha | V(x) | \Psi_\alpha \rangle = \int_{-\infty}^{+\infty} dx \sqrt{\frac{\alpha}{\pi}} e^{-\alpha x^2} V(x)$$

$$\rightarrow \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} dx V(x) \quad \text{for } \alpha \gg d^{-1}$$

$$= -\sqrt{\alpha} C \quad \text{with}$$

$$C = -\frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} dx |V(x)| \quad C < 0$$

$$E_\alpha \rightarrow \frac{\frac{p^2}{2m} \alpha}{4m} - \sqrt{\alpha} C < 0 \quad \text{for } \alpha \text{ sufficiently small}$$

$$E_0 \leq \langle \Psi_0 | H | \Psi_0 \rangle$$

$$\therefore E_0 < 0$$

$\therefore$ there is at least one bound state.